METRIC

A-A-56021
31 January 1991
SUPERSEDING
MIL-P-55010A(MI)
16 February 1965

COMMERCIAL ITEM DESCRIPTION

PLASTIC SHEET, POLYETHYLENE, TEREPHTHALATE (FOR CARTOGRAPHIC OPERATIONS)

The General Services Administration has authorized the use of this Commercial Item Description in preference to MIL-P-55010.

Abstract. This commercial item description covers a transparent, colorless polyethylene terephthalate plastic in sheet form. The plastic material is intended for use in military cartographic operations. The plastic sheet may be used for glazing which is exposed only intermittently to sunlight.

Salient characteristics. The material shall consist of clear polyethylene terephthalate plastic sheet without the addition of dyes, pigments, filler, or bonding agents. The material shall be free from opalescence and noticeable coloring when viewed at an angle of 90° to the plane of the material. The material shall transmit not less than 90 percent of incident white light when tested by means of a photoelectric densitometer. The physical requirements are as shown in table I.

TABLE I. Physical requirements.

	Maximum	Minimum	Test Method
Specific gravity, 25° Celsius (C)	1.45	1.35	ASTM D 792
Tensile strength, kpa	---	110240	ASTM D 882
Elongation at rupture, percent	---	60	ASTM D 621
Tear resistance, gr/cm	---	107160	ASTM D 1004
Water absorption, 24-hour immersion, %	0.7	---	ASTM D 570
Burning rate, centimeters per minute	12.0	---	ASTM D 568

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any data which may improve this document should be sent to: Commander, U.S. Army Missile Command, ATTN: AMSMI-RD-SE-TD-ST, Redstone Arsenal, AL 35898-5270.

AMSC N/A FSC 9330

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:58Z
Check the source to verify that this is the current version before use.

Contractor certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this Commercial Item Description, and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices. The government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Metric products. Products manufactured to metric dimensions will be considered on an equal basis with those manufactured using inch-pound units, provided they fall within specified tolerances using conversion tables contained in the latest revision of FED-STD-376, and all other requirements of this Commercial Item Description are met. If a product is manufactured to metric dimensions and those dimensions exceed the tolerances specified in the inch/pound units, a request should be made to the contracting officer to determine if the product is acceptable. The contracting officer has the option of accepting or rejecting the product.

Regulatory requirements. The offeror/contractor is encouraged to use recovered materials IAW Public Law 94-580 to the maximum extent practical.

Preservation, packaging, packing, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

CID-based part identification number. The following part identification number procedure is for government purposes and does not constitute a requirement for the contractor.

A	56021 - 1	Example of part identification number, A56021-1
		Dash number
		CID number
		Designates a CID

Notes:

a. Types. The purchaser should specify the type in the contract or purchase order. The plastic sheets shall be of the following types, based on thickness.

Type I	0.00075 millimeter (mm)
Type II	0.00125 millimeter
Type III	0.00186 millimeter

b. Dimensions. The purchaser should specify the dimensions in the contract or purchase order.

(1) Length and width. The material shall be furnished in the lengths and widths specified. A tolerance of $\pm$ 0.03125 mm of the length and width specified shall be allowed.

(2) Thickness. Unless otherwise specified, the thickness shall be 0.00186 $\pm$ 0.0002 mm, and the difference between the thickest and thinnest portions of any sheet shall be not greater than

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:58Z
Check the source to verify that this is the current version before use.

0.00015 mm. This material is also available in 0.00075 and 0.00125 mm thicknesses. The thickness and square yards per pound are as follows:

Thickness mm	Weight Square meter/kilogram
0.00075	9.46
0.00125	5.70
0.00186	3.76

(3) Dimensional stability. The material shall be subjected to a thermal dimensional stability test. The dimensional change of the material under changes in temperature and relative humidity shall be not greater than 0.0025 mm and the differential dimensional change of the material shall be not greater than 0.00075 mm.

(4) Thermal dimensional stability test.

(a) Specimens. Two 7.62 cm by 30.48 cm specimens shall be cut from each sheet of the sample. One specimen shall be cut from the lengthwise dimension of the sheet and the other from the crosswise dimension of the sheet. The specimens shall be index marked at a distance of 25.40 cm, ± 0.00127 cm, between the marked points.

(b) Procedure. The marked specimens shall be conditioned for not less than 24 hours at each of the following temperatures at a relative humidity of 50 percent, ± 2 percent: 15.5°, ± 1°C; 43.3°, ± 1°C; and 15.5°, ± 1°C in the order given. The distance between the index marks shall be measured to within 0.00127 cm at the end of each of the three 24-hour conditioning periods without removing the specimens from the conditioning temperatures. The difference between the average distance between the index marks measured at 15.5°C and the distance between the marks measured at 43.3° shall be considered the thermal dimension change. The smaller difference so found for each pair of specimens from each test sheet shall be subtracted from the larger difference found for the pair to determine compliance with the differential dimension change. Noncompliance shall constitute failure of this test.

(5) Source for referenced documents. FED-STD-376, "Preferred Metric Units for General Use By The Federal Government," may be obtained by sending a request for copies to the Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094. Copies of referenced ASTMs are available from the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103:

ASTM D 568 -	Standard Test Method for Rate of Burning and/or Extent and Time of Burning of Flexible Plastics in a Vertical Position
ASTM D 570 -	Standard Test Method for Water Absorption of Plastics
ASTM D 621 -	Standard Test Methods for Deformation of Plastics Under Load
ASTM D 792 -	Standard Test Methods for Specific Gravity (Relative Density) and Density of Plastics by Displacement
ASTM D 882 -	Standard Test Methods for Tensile Properties of Thin Plastic Sheeting

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:58Z
Check the source to verify that this is the current version before use.

ASTM D 1004- Standard Test Method for Initial Tear Resistance of Plastic Film and Sheeting

MILITARY INTERESTS:

Custodian:
Army - MI
Air Force - 99
Navy - YD

Review Activities:
Army - MD, MR
DLA - GS
GSA - 7FXEG
NASA - MSF

User Activities:
Army - ER, GL
Navy - CG, MC

PREPARING ACTIVITY:
Army - MI

Project No. 9330-1148

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T09:58Z
Check the source to verify that this is the current version before use.